

Riley Brown

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EDUCATION

University of Nebraska-Lincoln

B.S. Mathematics and Data Science, Minor in Computer Science, Lincoln, NE | Aug 2023 - May 2027 (expected)

- GPA 3.92/4.00, University Honors Program

Budapest Semesters in Mathematics

Study abroad, mathematics, Budapest, Hungary | Jan 2026 - Jun 2026

PREPRINTS

Fixed-Parameter Tractability of t -Uniform Hypergraphicality

Riley Brown, Istvan Miklos

arXiv preprint, 2026

arXiv

RESEARCH EXPERIENCE

Undergraduate Researcher - Quantum LDPC Codes

University of Nebraska-Lincoln, Lincoln, NE | Fall 2026

- Testing whether collision-free geodesic-flow operators on Ramanujan complexes can sharpen iterative-decoder thresholds for quantum LDPC codes

Research Assistant, pseudorandomness

North Carolina State University, Raleigh, NC | Summer 2026

- Studied Ta-Shma's epsilon-balanced code construction with Prof. Chin Ho Lee as a research partner
- Found simpler constructions and minor improvements to the Hsieh-Mohanty-Zheng construction

Research Assistant, t -uniform hypergraph degree sequences and FPT algorithms

Budapest Semesters in Mathematics, Budapest, Hungary | Spring 2026

- Proved t -uniform hypergraphicality is FPT in (k, t) for compressed degree sequences, extending known bounded-range results
- Reduced realizability to integer programming feasibility via bounded-dimension spectra and balancing hinge-flips

Summer Researcher, algorithm design

Harvey Mudd College, Claremont, CA | Summer 2025

- Improved exponential-time algorithms for sumset recognition, an NP-complete problem in additive combinatorics (team of four)
- Sharpened branching strategies and base-case parameters for graph-theoretic reductions

PROJECTS

leanverify

Verification kernel for an LLM-driven proof loop: checks a Lean 4 proof against Mathlib, rejecting sorries and smuggled axioms | python, lean 4 | 2026

SKILLS

AI for mathematics: Lean 4 and Mathlib verification tooling, LLM-assisted proof search, agentic research harnesses

Languages: Python, MATLAB, Java, SQL, LaTeX

ACTIVITIES

Putnam Exam Practice, Quiz Bowl Club, Directed Reading Program